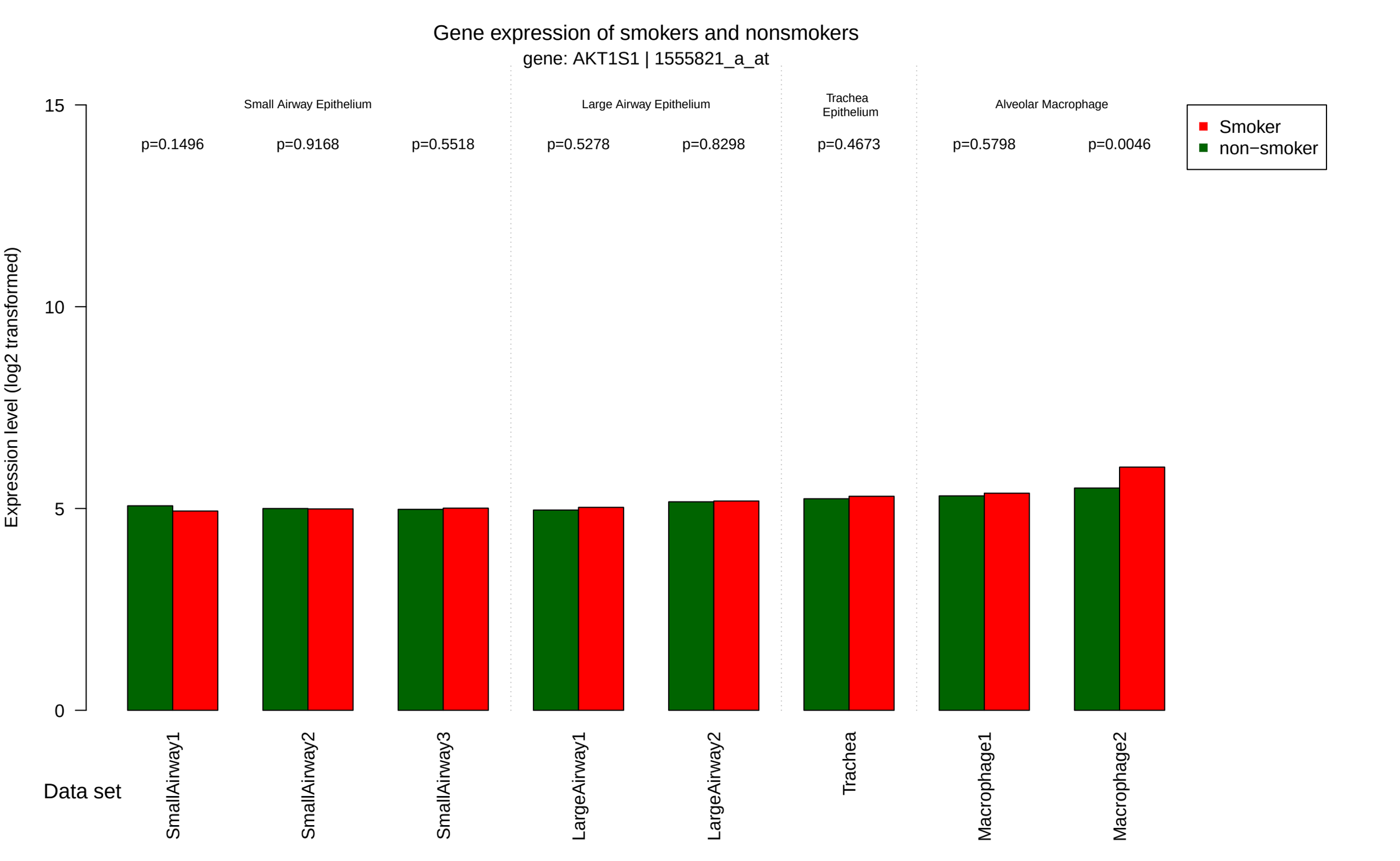
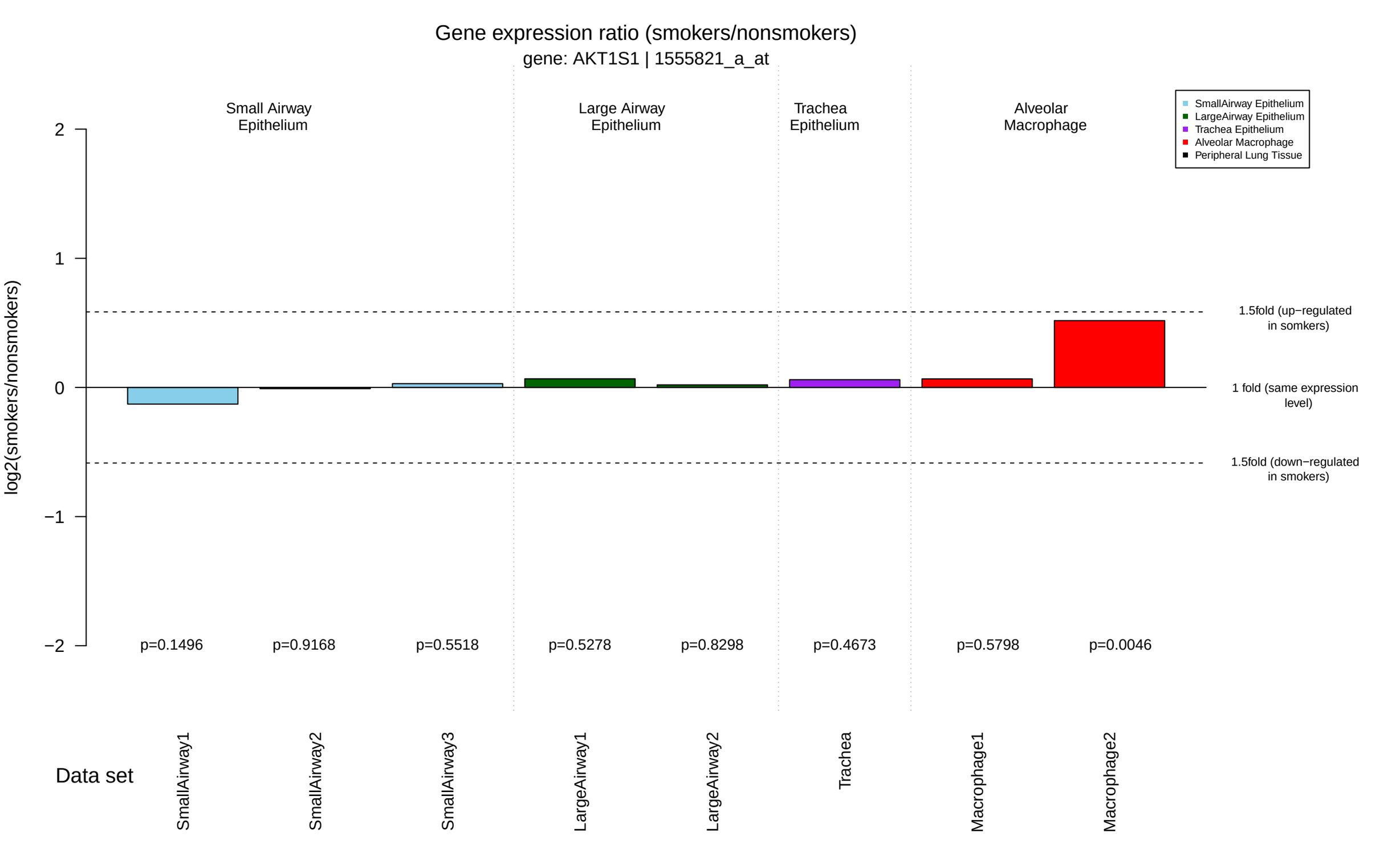
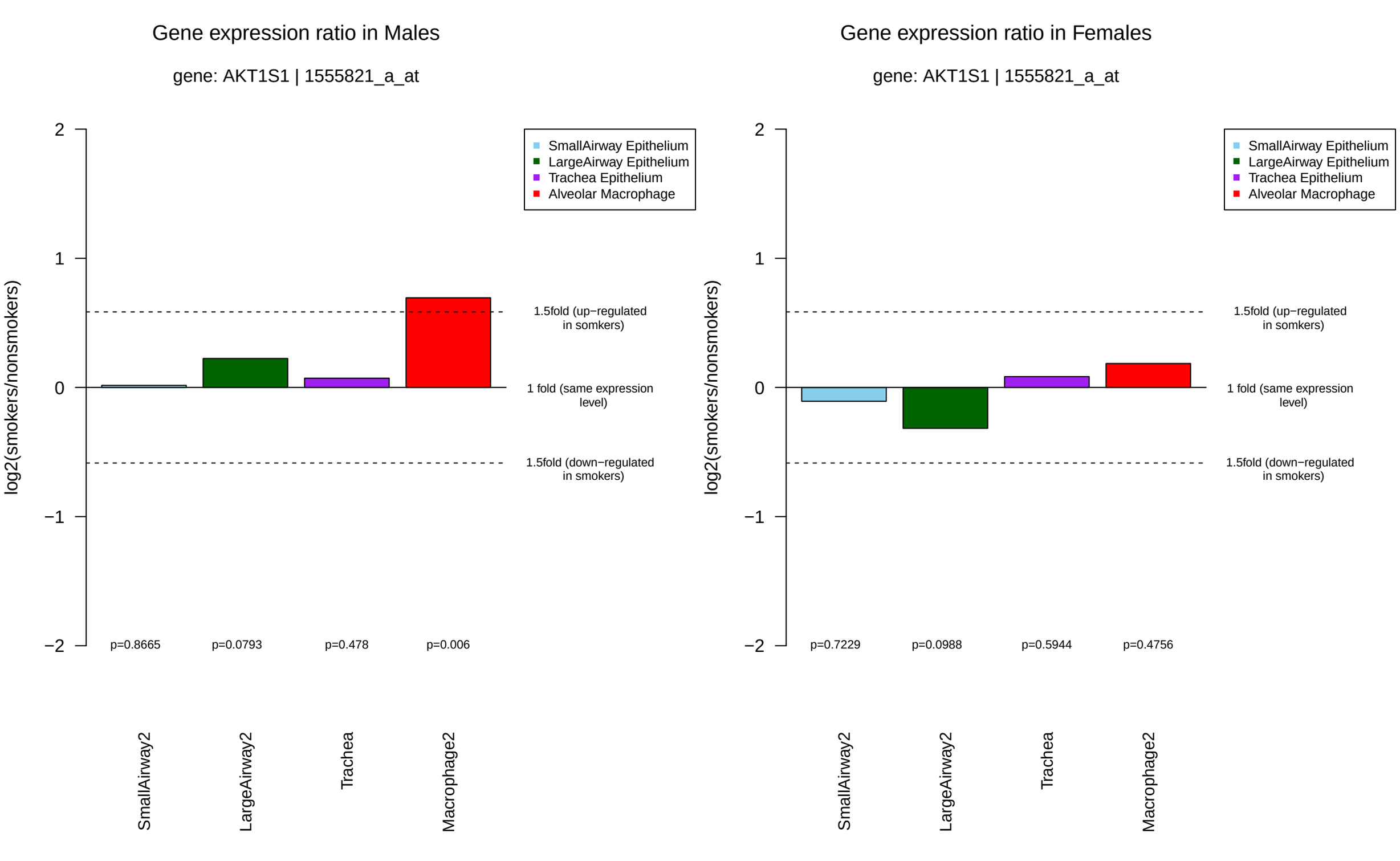
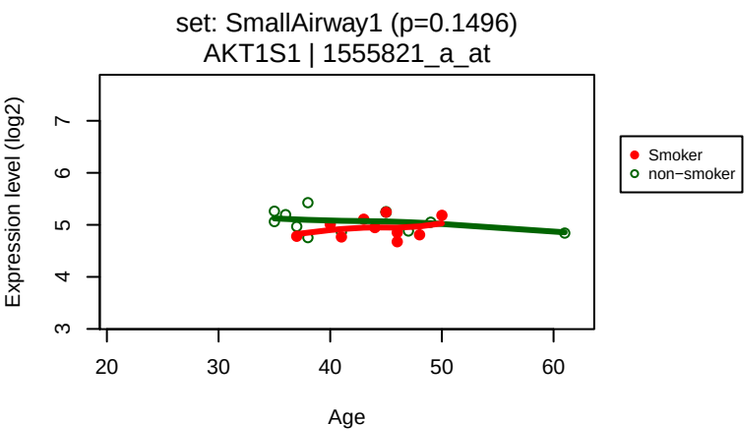
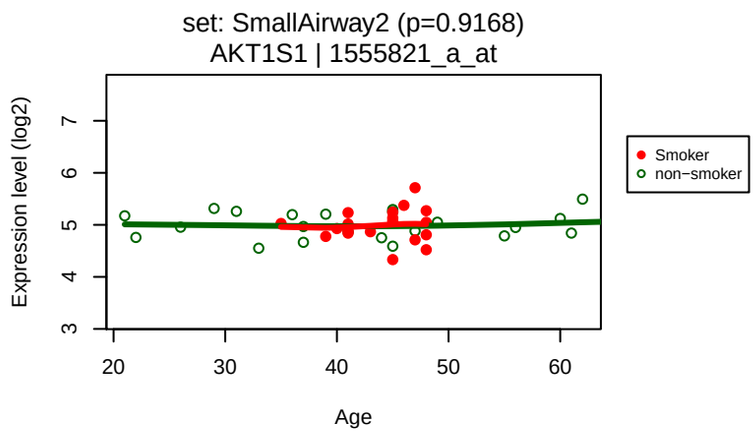
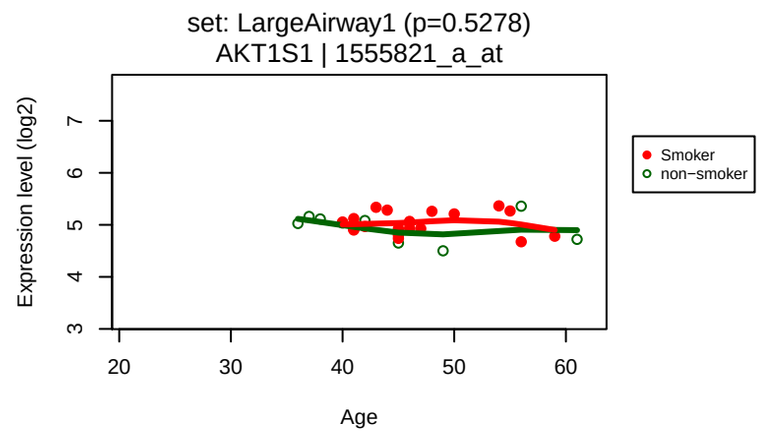

Gene expression of smokers and nonsmokers
gene: AKT1S1 | 1555821_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1496
p=0.9168
p=0.5518
p=0.5278
p=0.8298
p=0.4673
p=0.5798
p=0.0046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

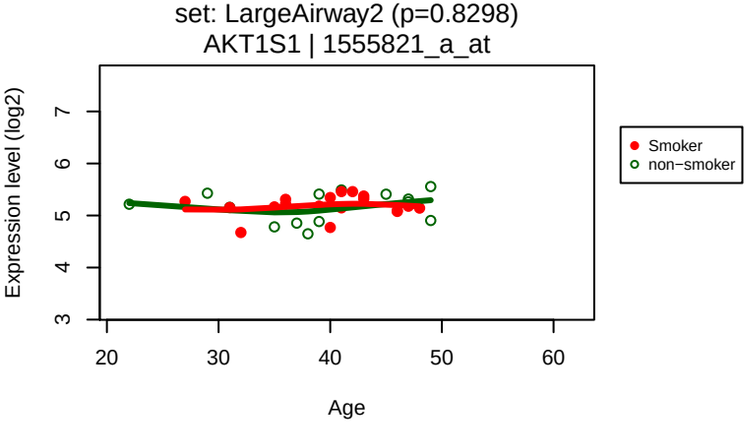

Gene expression ratio (smokers/nonsmokers)
gene: AKT1S1 | 1555821_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1496
p=0.9168
p=0.5518
p=0.5278
p=0.8298
p=0.4673
p=0.5798
p=0.0046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

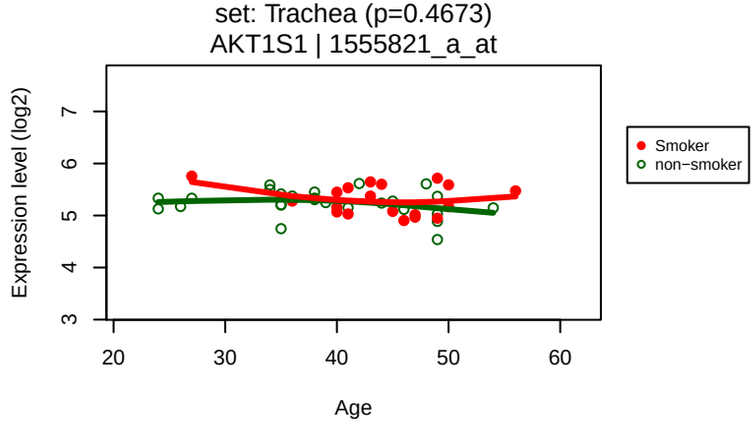

Gene expression ratio in Males
gene: AKT1S1 | 1555821_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8665
p=0.0793
p=0.478
p=0.006
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT1S1 | 1555821_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7229
p=0.0988
p=0.5944
p=0.4756
SmallAirway2
LargeAirway2
Trachea
Macrophage2

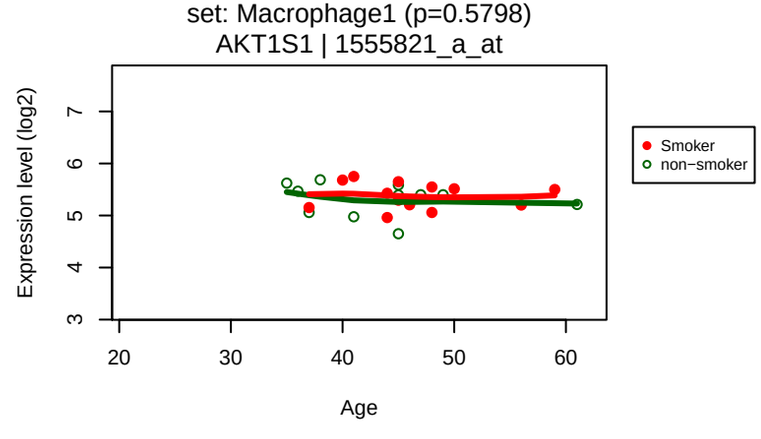

set: SmallAirway1 (p=0.1496)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker

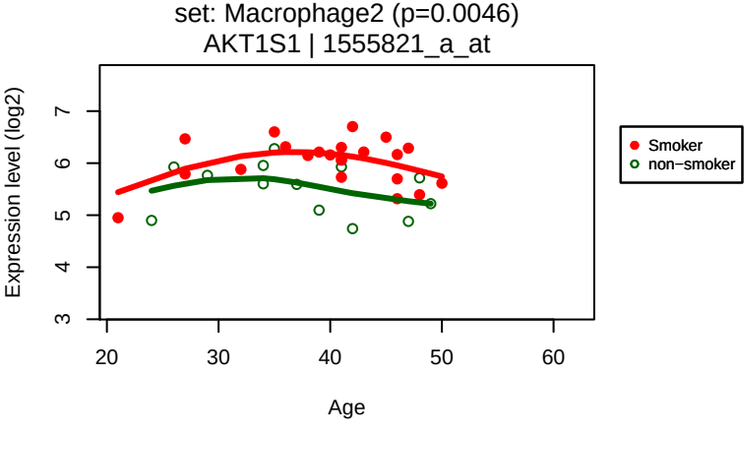

set: SmallAirway2 (p=0.9168)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker

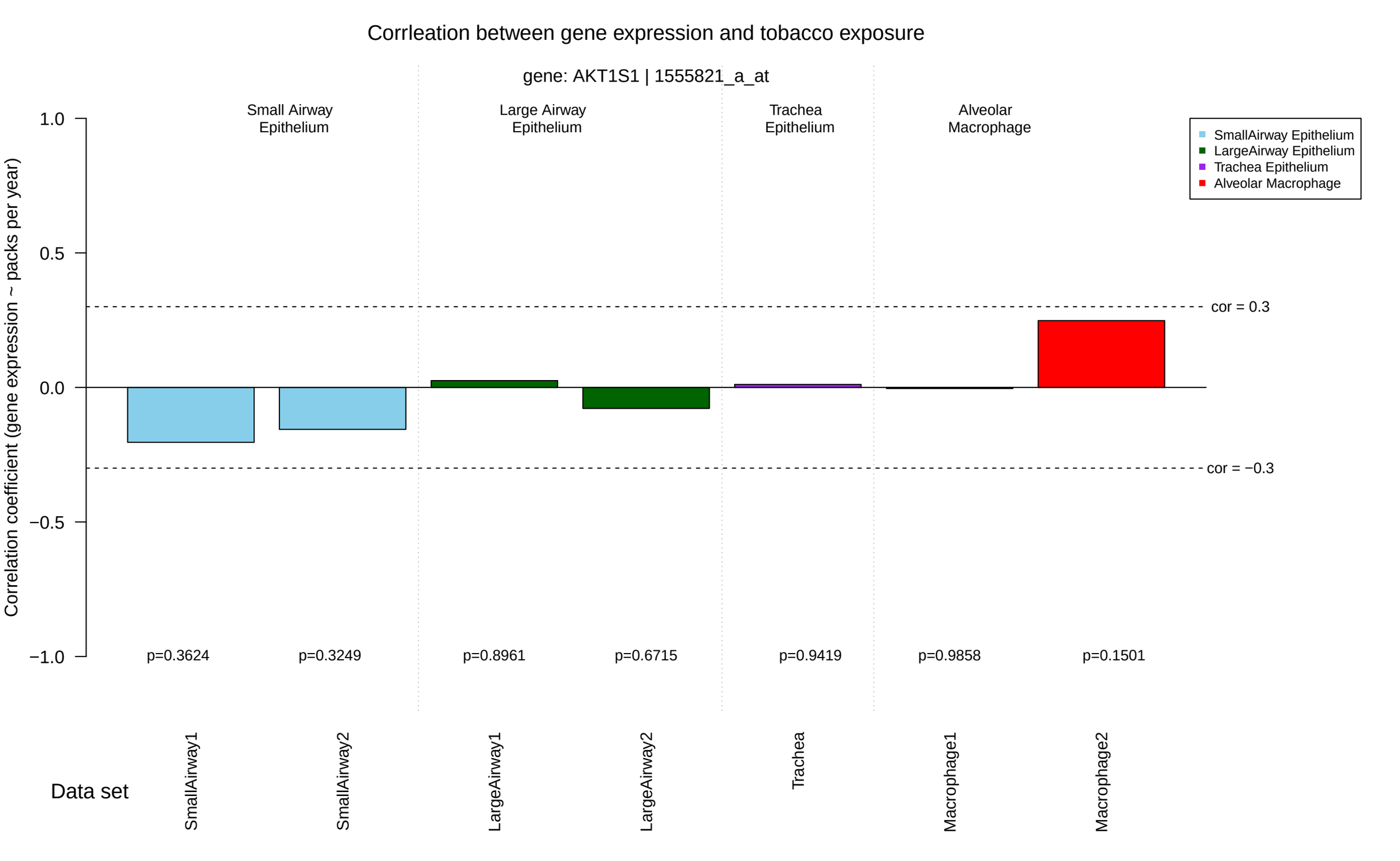

set: LargeAirway1 (p=0.5278)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8298)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4673)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5798)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0046)
AKT1S1 | 1555821_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT1S1 | 1555821_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3624
p=0.3249
p=0.8961
p=0.6715
p=0.9419
p=0.9858
p=0.1501
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2